

NIKHIL SAHU

Cybersecurity and Digital Forensics | Final Year B.Tech

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PROFESSIONAL SUMMARY

Final-year B.Tech student in Cybersecurity and Digital Forensics with hands-on experience in penetration testing, Vulnerability Assessment and Penetration Testing (VAPT), and web application security. Led a 4-member VAPT team during a 6-month engagement at Codevirus Security and possess 3+ years of bug bounty hunting experience across 10+ HackerOne programs, including a responsibly disclosed vulnerability acknowledged by Quick Heal Technologies. Co-developed an AI-based zero-day malware detection framework and hold an Indian Patent. Seeking opportunities in web application penetration testing, VAPT, and offensive security.

Responsible Disclosure | Quick Heal Technologies

Identified and responsibly disclosed a brute force vulnerability in Quick Heal's product key validation system, acknowledged and patched by their security team.

PROFESSIONAL EXPERIENCE

Cyber Security Professional (Team Lead) | Codevirus Security

January 2026 - July 2026 | Remote

- Led a 4-member VAPT team conducting penetration testing on Codevirus Security's web applications; managed task delegation, daily standups, and final report delivery.
- Performed black-box and grey-box assessments, identifying vulnerabilities including Authentication Bypass, OTP Bypass, CSRF, Session Hijacking, Brute Force, and Information Disclosure.
- Documented findings using CVSS v3.1 severity scoring and provided remediation recommendations adopted by clients.

Cyber Security Intern | Samatrix Consulting Pvt. Ltd.

August 2025 | Indore

- Conducted malware analysis in a controlled sandbox environment, identifying malicious behavior patterns and classifying threat types.
- Assisted in designing cybersecurity defense mechanisms and contributed to threat detection activities for client-facing security projects.

Security Researcher - Bug Bounty Hunter | HackerOne

2021 - 2024 | Remote

- Participated in 10+ bug bounty programs over 3+ years, identifying and reporting authentication bypasses, misconfigurations, and insecure API implementations across live production systems.
- Performed full-cycle reconnaissance, endpoint enumeration, and manual exploitation using Burp Suite and Linux-based tools.

Cyber Security Trainee | AlanceSec

April 2024 - Jun 2024 | Remote

- Completed intensive hands-on training in vulnerability assessment, exploitation techniques, and penetration testing methodologies.
- Performed web application testing using Burp Suite, Metasploit, and Nmap, identifying and exploiting common OWASP Top 10 vulnerabilities.

EDUCATION

B.Tech - Computer Technology (Cyber Security and Digital Forensics)

SAGE University Indore | CGPA: 7.82 / 10 (up to 6th Semester) | 2023 - 2027 | Indore, India

TECHNICAL SKILLS

Offensive Security: Vulnerability Assessment and Penetration Testing (VAPT), Penetration Testing, Bug Bounty Hunting, Reconnaissance, Vulnerability Research, Exploit Development.

Web Application Security: OWASP Top 10, SQL Injection, XSS, IDOR, Authentication Bypass, Session Analysis.

Security Tools: Burp Suite, Nmap, Metasploit, Wireshark, Nessus, Kali Linux.

Malware Analysis: Dynamic Analysis, Behavior Pattern Detection, Sandboxing.

Development: Python, JavaScript, HTML, CSS, SQL, Bash Scripting, Flask.

Systems and Networking: Linux, Command Line, TCP/IP, Network Protocols, Packet Analysis.

PROJECTS

Windows Persistence and Red Team Simulation | *Reverse Shell, Windows Shell Scripting*

Simulated real-world red team operations in a controlled lab, implementing Windows persistence techniques, reverse shell mechanisms, and registry-based startup methods.

Cryptographic Hash Generator | *Python, Cryptography*

Python-based hash generator supporting MD5, SHA-256, SHA3-224, BLAKE2s, and SHAKE-128 for file integrity verification.

GitHub: github.com/heynick1337/hashgen

Web Authentication System | *Python, Flask, HTML, CSS, JavaScript, MySQL*

Built a secure web authentication system with user validation, session management, and access control following OWASP principles.

GitHub: github.com/heynick1337/AuthShield

Bug Bounty Recon Framework | *Python, Flask, HTML, CSS, JavaScript*

Web-based recon dashboard for bug bounty hunting. Features subdomain enumeration, port scanning, security header analysis, and tech stack detection with report export.

GitHub: github.com/heynick1337/recon-framework

PATENTS AND RESEARCH WORK

Adaptive AI-Based Zero-Day Malware Detection and Neutralization Device | *February 2026*

SAGE University Indore - Designed an AI-driven cybersecurity framework for real-time zero-day malware detection, autonomous threat response, and adaptive protection across enterprise and IoT environments.

System and Method for AI-Based Community Weather Forecasting | *Indian Patent* | *February 2025*

SAGE University Indore - Developed an AI-based localized weather forecasting system using machine learning algorithms. Filed and published through the Indian Patent Office.

HONORS & AWARDS

Certificate of Appreciation | *Codevirus Security* | *July 2026*

Certificate ID: CVS/INT/2026/003 | Awarded in recognition of dedication, hard work, and valuable contributions during the Cyber Security Internship at Codevirus Security.

CERTIFICATIONS

Masters in Cybersecurity and Penetration Testing | *Jetking Indore* | *AlanceSec Mumbai* | *September 2024*

Serial No: 2324-006485 | **Grade:** A | **288 Hours** | Covered ethical hacking, web application penetration testing, vulnerability assessment, and exploitation techniques using Burp Suite, Metasploit, Nmap, and Kali Linux.

Cyber Security Analyst Internship Training | *AlanceSec*

Certificate No: ALN225 | 3-month training program covering cybersecurity analysis, threat detection, and security operations. ISO 27001 Certified program.

Cyber Security Internship Completion | *Samatrix Consulting Pvt. Ltd.* | *September 2025*

4-week internship in Cyber Security including malware analysis and cybersecurity defense mechanism design.

Cyber Security Internship and Professional Experience | *Codevirus Security* | *July 2026*

Certificate ID: CVS-CS-2026-036 | 6-month engagement in penetration testing and security assessment.